

## Module 4: Vector Spaces

4.1

Def. of a Group: Let  $G$  be a non-empty

set together with one binary operation ' $\circ$ '. Then  $(G, \circ)$  is called a group if it satisfies the following:

(i) Associativity:  $a \circ (b \circ c) = (a \circ b) \circ c$   
for  $a, b, c \in G$ ,

(ii) Existence of identity: For all  $a \in G$ ,

there exists an element  $e \in G$  such that  $a \circ e = e \circ a = a$ .

Here  $e$  is called an identity element of  $G$  with respect to ' $\circ$ '.

(iii) Existence of inverse: For each  $a \in G$ ,

there exists an element  $b \in G$  such that  $a \circ b = b \circ a = e$ .

Here  $b$  is called an inverse of  $a$  in  $G$  with respect to ' $\circ$ '.

Note: A mapping  $\circ : G \times G \rightarrow G$  is called a binary operation on  $G$ .

E.g. (i)  $(\mathbb{N}, +)$  is not a group.

for,  $0 \notin \mathbb{N}$  ( $0$  is an additive identity.)

(ii)  $(\mathbb{Z}, +)$  is a group.

(iii)  $(\mathbb{Z}, \cdot)$  is not a group.

for,  $2 \in \mathbb{Z}$  but  $\frac{1}{2} \notin \mathbb{Z}$

i.e., multiplicative inverse of 2 is not in  $\mathbb{Z}$ .

(iv)  $(\mathbb{R}, +)$  is a group.

(v)  $(\mathbb{R}, \cdot)$  is not a group.

for,  $0 \in \mathbb{R}$  has no multiplicative inverse.

(vi)  $(\mathbb{R}^*, \cdot)$  is a group, where  $\mathbb{R}^* = \mathbb{R} - \{0\}$ .

Abelian Group: A group  $(G, \cdot)$  is said to

be abelian if  $a \cdot b = b \cdot a$  for all  $a, b \in G$ .

e.g. (i)  $(\mathbb{Z}, +)$ ,  $(\mathbb{R}, +)$ ,  $(\mathbb{Q}, +)$  are all abelian groups.

(ii)  $(\mathbb{R}^*, \cdot)$  is an abelian group.

Def. of a Ring: Let  $R$  be a non-empty set together with two binary operations '+' and '•'. Then  $(R, +, \cdot)$  is called a ring if

- (i)  $(R, +)$  is an abelian group
- (ii)  $(R, \cdot)$  is a semi group  
(i.e.,  $a \cdot (b \cdot c) = (a \cdot b) \cdot c$  for all  $a, b, c \in R$ )
- (iii) Distributive laws hold good in  $R$ .  
i.e.,  $a \cdot (b + c) = a \cdot b + a \cdot c$   
and  $(a + b) \cdot c = a \cdot c + b \cdot c$   
for all  $a, b, c \in R$ .

e.g.:

- (i)  $(\mathbb{Z}, +, \cdot)$  is a ring
- (ii)  $(\mathbb{R}, +, \cdot)$  is a ring
- (iii)  $(\mathbb{Q}, +, \cdot)$ ,  $(\mathbb{C}, +, \cdot)$  are all rings.

Definition of a Field: Let  $F$  be a non-empty set with two binary operations '+' and '•'. Then  $(F, +, \cdot)$  is called a field if

- (i)  $(F, +)$  is an abelian group
- (ii)  $(F^*, \cdot)$  is an abelian group ( $F^* = F - \{0\}$ )
- (iii) Distributive laws are hold good in  $F$



i.e.,  $a \cdot (b+c) = a \cdot b + a \cdot c$   
and  $(a+b) \cdot c = a \cdot c + b \cdot c$  for all  $a, b, c \in F$ .

e.g. (i)  $(\mathbb{Z}, +, \cdot)$  is not a field.

(ii)  $(\mathbb{Q}, +, \cdot)$  is a field

(iii)  $(\mathbb{R}, +, \cdot)$  and  $(\mathbb{C}, +, \cdot)$  are fields.

Vector Space: Let  $V$  be a non-empty set whose elements are called vectors and  $F$  be a field whose elements are called scalars. Then  $(V, +, \cdot)$  is said to be a vector space over  $F$  if

(i)  $(V, +)$  is an abelian group.

(ii)  $a \in F, \alpha \in V \Rightarrow a \cdot \alpha \in V$  ( $\forall a \in F, \alpha \in V$ )

(iii)  $a \cdot (\alpha + \beta) = a \cdot \alpha + a \cdot \beta$  for  $a \in F$  and  $\alpha, \beta \in V$

(iv)  $(a+b) \cdot \alpha = a \cdot \alpha + b \cdot \alpha$  for  $a, b \in F$  and  $\alpha \in V$

(v)  $a \cdot (b \cdot \alpha) = (a \cdot b) \cdot \alpha$  for  $a, b \in F$  and  $\alpha \in V$

and (vi)  $1 \cdot \alpha = \alpha$  for  $1 \in F$  and  $\alpha \in V$ .

Note:

If  $(V, +, \cdot)$  is a vector space over a field  $F$ , then we say  $V(F)$  is a vector space (or, a linear space)

eg. (i)  $\mathbb{R}$  is a vector space over a field  $(\mathbb{Q}, +, \cdot)$   
i.e.,  $\mathbb{R}(\mathbb{Q})$  is a vector space

(ii)  $\mathbb{C}(\mathbb{R})$  is a vector space

(iii)  $\mathbb{R}(\mathbb{C})$  is not a vector space.

for,  $1 \in \mathbb{R}$  and  $i \in \mathbb{C} \Rightarrow 1 \cdot i = i \notin \mathbb{R}$

(iv)  $\mathbb{R}^n(\mathbb{R})$  is a vector space ( $n=1, 2, 3, \dots$ )

(v)  $(M_{2 \times 2}, +, \cdot)$  is a vector space

over a field  $(\mathbb{R}, +, \cdot)$  (here  
 $M_{2 \times 2} = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in \mathbb{R} \right\}$ )

Subspace: Let  $V(F)$  be a vector space.

A non-empty subset  $W$  of  $V$  is ~~called~~  
said to be a subspace of  $V$  over a field  $F$  if

$W$  is itself a vector space over  $F$  under  
the same operations as defined on  $V$ .

Example ① Clearly

$W = \{ (a, b, c) \mid a, b, c \in \mathbb{R} \}$  is a subspace  
of a vector space  $\mathbb{R}^3(\mathbb{R})$ .

②  $W = \left\{ \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$  is a  
a vector space

Subspace of  $M_{2 \times 2}(\mathbb{R})$ .



\* The necessary and sufficient condition for a non-empty subset  $W$  of a vector space  $V(F)$  to be a subspace of  $V$  is

$$\alpha, \beta \in W \Rightarrow \alpha + \beta \in W.$$

Note: ① The intersection of any two subspaces of a vector space is a subspace.

② The intersection of an arbitrary collection of subspaces of a vector space is also a subspace.

③ The union of two subspaces of a vector space is not necessarily a subspace.

④ The union of two subspaces  $W_1$  and  $W_2$  of a vector space  $V(F)$  is a subspace of  $V$  if and only if  $W_1 \subseteq W_2$  or  $W_2 \subseteq W_1$ .

Def.: Let  $W_1$  and  $W_2$  be two subspaces of a vector space  $V(F)$ , then  $W_1 + W_2 = \{ \alpha + \beta \mid \alpha \in W_1, \beta \in W_2 \}$  is called the linear sum of  $W_1$  and  $W_2$ .

Note:  $W_1 + W_2$  is a subspace of  $V(F)$ .

Def.: Let  $W_1$  and  $W_2$  be two subspaces of a vector space  $V(F)$ . Then  $V$  is said to be a

direct sum of  $w_1$  and  $w_2$  if each element of  $V$  can be uniquely expressed as the sum of an element of  $w_1$  and an element of  $w_2$ .

If  $V$  is a direct sum of  $w_1$  and  $w_2$ , then we write  $V = w_1 \oplus w_2$

\* The necessary and sufficient condition for a vector space  $V$  to be the direct sum of its subspaces  $w_1$  and  $w_2$  are

$$(i) V = w_1 + w_2 \quad (ii) w_1 \cap w_2 = \{0\}.$$

### Linear combination of vectors

Def.: Let  $V(F)$  be a vector space. An element  $\alpha \in V$  is said to be a linear combination of  $\alpha_1, \alpha_2, \dots, \alpha_n \in V$  if there exists  $a_1, a_2, \dots, a_n \in F$  such that  $\alpha = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n$ .

Example: ① The vector  $\alpha = (-7, -6)$  is a linear combination of the vectors  $\alpha_1 = (-2, 3)$

and  $\alpha_2 = (1, 4)$   $\left( \begin{array}{l} \text{Since } \alpha \in \text{span}\{\alpha_1, \alpha_2\} \\ \text{then} \end{array} \right)$

$$\text{since } (-7, -6) = 2(-2, 3) - 3(1, 4)$$

$$\text{i.e., } \alpha = 2\alpha_1 + (-3)\alpha_2$$

Linear Span: Let  $V(F)$  be a vector space and  $S$  be any non-empty subset of  $V$ . Then the linear span of  $S$  (or, simply span of  $S$ ) is denoted by  $L(S)$  (or,  $\text{span}(S)$ ) and defined as the set of all finite linear combinations of elements of  $S$ .

$$\text{i.e., } L(S) = \left\{ \sum_{i=1}^n a_i \alpha_i \mid a_i \in F, \alpha_i \in S \text{ for } i=1,2,\dots,n \right\}.$$

Clearly  $S \subseteq L(S)$ .

Note: ①  $\text{span}(S)$  is the smallest subspace of  $V$  that contains  $S$ .

Examples: ①  $L(\{(1,0,0), (0,1,0), (0,0,1)\}) = \mathbb{R}^3(\mathbb{R}).$

for, for  $(x,y,z) \in \mathbb{R}^3$ ,  $(x,y,z) = x(1,0,0) + y(0,1,0) + z(0,0,1).$

②  $L(\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}) = M_{2 \times 2}(\mathbb{R}).$

for, for  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2 \times 2}$ ,  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$



③ Show that the set  $S = \{(2, 5), (1, 3)\}$  is a spanning set of  $\mathbb{R}^2$  ( $\mathbb{R}$ )

sol: let  $(a, b) \in \mathbb{R}^2$ . Suppose  $(a, b) = r(2, 5) + s(1, 3)$

$$\text{Then } 2r + s = a$$

$$5r + 3s = b$$

$$\text{i.e., } \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} r \\ s \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\xrightarrow{2R_2 - 5R_1} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ s \end{bmatrix} = \begin{bmatrix} a \\ 2b - 5a \end{bmatrix}$$

$$\text{which gives } 2r + s = a$$

$$\text{and } s = 2b - 5a$$

$$\Rightarrow r = \frac{6a - 2b}{2} \Rightarrow r = 3a - b$$

$$\therefore (a, b) = (3a - b)(2, 5) + (2b - 5a)(1, 3).$$

and hence every element of  $\mathbb{R}^2$  can be

expressed ~~as a~~ linear combination of

~~elements~~ of  $S$ . Thus  $L(S) = \mathbb{R}^2$

Linearly dependent and independent set of vectors.

Def: Let  $V(F)$  be a vector space over a field  $F$ .

Then a finite set  $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$  of vectors in  $V$  is said to be (i) linearly dependent if there exists scalars  $a_1, a_2, \dots, a_n \in F$

not all are zero such that

$$a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = 0$$

(ii) linearly independent if  $a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n = 0$   
where  $a_1, a_2, \dots, a_n \in F$   
implies  $a_1 = 0, a_2 = 0, \dots, a_n = 0$

Example: ① The set of vectors  $(2, 1, 4), (1, -1, 2), (3, 1, -2)$  i.e.,  $\{(2, 1, 4), (1, -1, 2), (3, 1, -2)\}$  is a linearly independent subset of  $\mathbb{R}^3(\mathbb{R})$ .

Suppose  $a, b, c \in \mathbb{R}$  such that

$$a(2, 1, 4) + b(1, -1, 2) + c(3, 1, -2) = 0$$

$$\Rightarrow 2a + b + 3c = 0; a - b + c = 0; 4a + 2b - 2c = 0$$

$$\text{i.e., } \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 4 & 2 & -2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow[R_3 - 2R_1]{R_2 - 2R_1} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -4 & -1 \\ 0 & 0 & -8 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore, we get  $a = 0, b = 0, c = 0$

Hence the vectors  $(2, 1, 4), (1, -1, 2), (3, 1, -2)$  are linearly independent.

② Show that the set  $S = \{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$  is a linearly dependent subset of  $\mathbb{R}^3(\mathbb{R})$ .

Note: If every finite subset of an infinite set  $S$  of vectors is linearly independent, then  $S$  is linearly independent.

(2) If  $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$  is a linearly independent subset of a vector space  $V(F)$ , then every element  $\alpha \in L(S)$  has a unique representation in the form  $a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n$ , where  $a_1, a_2, \dots, a_n \in F$ .

Basis of a vector space:

Definition: Let  $V(F)$  be a vector space and  $S$  any non-empty subset of  $V$ . Then  $S$  is said to be basis of  $V$  if (i)  $(\text{all } \mathbb{R})$   $S$  is linearly independent  
(ii)  $L(S) = V$

Examples: (1)  $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

is linearly independent and  $L(S) = \mathbb{R}^3(\mathbb{R})$ .

$\therefore S$  is a basis of  $\mathbb{R}^3(\mathbb{R})$ .

This basis is called ~~the~~ standard basis of  $\mathbb{R}^3(\mathbb{R})$ .

(2)  $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

is the standard basis of  $M_{2 \times 2}(\mathbb{R})$



③ verify the vectors  $(1, 2, 1)$ ,  $(3, 1, 5)$ ,  $(3, -4, 7)$  form a basis of  $\mathbb{R}^3(\mathbb{R})$  or not.

Sol: Let  $a, b, c \in \mathbb{R}$  such that

$$a(1, 2, 1) + b(3, 1, 5) + c(3, -4, 7) = 0$$

$$\begin{aligned}\text{Then } a + 3b + 3c &= 0 \\ 2a + b - 4c &= 0 \\ a + 5b + 7c &= 0\end{aligned}$$

$$\text{i.e., } \begin{bmatrix} 1 & 3 & 3 \\ 2 & 1 & -4 \\ 1 & 5 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - R_1}]{\phantom{R_2 - 2R_1}} \begin{bmatrix} 1 & 3 & 3 \\ 0 & -5 & -10 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow{5R_3 + 2R_2} \begin{bmatrix} 1 & 3 & 3 \\ 0 & -5 & -10 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore,  $a, b, c$  are all not zero ~~and hence~~.

Hence, the vectors  $(1, 2, 1)$ ,  $(3, 1, 5)$ ,  $(3, -4, 7)$  are linearly dependent.

Thus, the vectors  $(1, 2, 1)$ ,  $(3, 1, 5)$ ,  $(3, -4, 7)$  do not form a basis of  $\mathbb{R}^3(\mathbb{R})$ .

Dimension of a vector space:

Let  $V(F)$  be a vector space. The number of elements of a basis of  $V$  is called the dimension of  $V$  over  $F$  and we write  $\dim_F(V)$  or simply  $\dim(V)$ .